

MSE 2202B DESIGN DOCUMENT LAB 4 TEAM 5

SPECIFICATION AND PLANNING PHASE

Tasks Completed February 13, 2026

- Problem Statement Established
- Goals & Constraints Established
- Vehicle Subsystem Trade Studies Completed

Problem Statement:

Resource extraction operations in hazardous environments need a way to autonomously navigate unknown terrain to identify and retrieve high-value objects (erasers), especially where long distance communication is unreliable. Thus, the objective of this project is to design and build an autonomous mobile robot capable of collecting pencil erasers from a defined area, separating them by colour, and returning them to a base location within a fixed time limit.

Constraints & Ways to Measure:

ID	Constraints	How to Measure
1.	Size	The robot must be able to fit in the MSE lab locker.
2.	Sorting of Erasers	The picked-up erasures must be sorted either on the robot or on a stationary sorting system, and must be done based on wavelength (colour)
3.	Navigation	The robot must be able to traverse the area of operations without any hindrance or help from external sources.
4.	Power/Battery	The robot must be able to perform its entire task on a single charge of the batteries used, and the batteries used must not be modified.
5.	Weight Drivable	The robot must be able to move on its own weight and be light enough to be carried by a human (< 3 kg).
6.	Time-Efficient	Perform the entire operation within the limit of 2 minutes
7.	Autonomous	Operations must be performed without any human intervention or wireless communication.
8.	Budget	The overall cost of components for the robot, excluding components provided by the MSE 2202 lab kit, must not exceed \$50. The overall cost for FDM and laser cutting must not exceed \$20.
9.	Durability	The robot must be able to perform its task without the expense of its components. The robot must be able to perform the entire operation > 50 times without any need for component replacement.
10.	Manufacturability	The robot must be manufactured using available materials/components. This includes 4x MSE 2202 lab kit, FDM printed ABS/PETG/PLA components, and external components adhering to the budget constraint.

Objectives & Ways to measure:

ID	Objectives	How to Measure
1.	Simplicity of Design	The robot should be made with a minimum no. of components to increase reliability and durability
2.	Sorting Accuracy & Amount	Sort accuracy (the sorting between high-value and low-value erasers) should be > 85%.
3.	Efficient Weight Distribution	The robot should be balanced and must not stumble under its own weight.
4.	Navigation Accuracy	Should traverse the operation without losing sight of goals and have at least 90% success rate on test runs.
5.	Efficient UI	Clear indication of each process step and a failsafe in case of incidents or wrong performance of the task
6.	Budget	The total cost of external components should be < \$20 provided by the project.

Rough Concept Generation:

ID	Concept	Readiness (Keep/Drop)	Reasoning
Robot Movement			
1.	Gears with Tracks	Keep	
2.	Aftermarket Rubber Wheels	Keep	
3.	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	Keep	
Chassis Design			
1.	4 Wheel Drive	Keep	
2.	2 Wheel Drive	Keep	
	S Wheel Drive + Dedicated Steering	Drop	Too complex with limited benefits.
Eraser Sorting			
1.	Conveyor Belt + 2 Lane Knocking System	Keep	
2.	Steep Slide with Hatch System	Keep	
3.	Slide With Sorting Wheel	keep	
4.	Angle Slide with Sensor Knocking System	Drop	It requires very precise timing and multisystem cooperation.
Eraser Drop Mechanism			
1.	Slide	Keep	
2.	Conveyor Belt	Drop	Require too much power and components
3.	Gate	Keep	
Eraser Pick Up Sensing			
1.	Camera	Drop	ESP32 does not have enough computational resources for image processing.
2.	UltraSonic Sensor	Keep	
3.	Infrared Sensor	Keep	

4.	UltraSonic + Infrared Sensor	Keep	
Eraser Pick Up Mechanism			
1.	Scoop pick-up	keep	
2.	Sticky Conveyor Belt	Drop	The Stickiness might lose its effectiveness over time
3.	Claw	keep	
4.	Vacuum Suction	Drop	Require many additional components on the robot
5.	Stabbing needle arm	Drop	Requires too much precision, too slow
6.	Scoop pick-up + Gate	Keep	

Go/No-Go Matrix:

ID	Concepts	Size	Sorting of Erasers	Navigation	Power/Battery	Weight Drivable	Time-Efficient	Autonomous	Budget	Durability	Manufacturability	Reasoning
1	Gears with Tracks	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Go	Maybe	- Requires precise alignment, tension control, and additional mechanical components, increasing fabrication complexity. - Higher material cost reduces overall efficiency and mobility compared to simpler wheel-based systems.
2	Aftermarket Rubber Wheels	Go	N/A	Go	N/A	Go	Go	N/A	Maybe	Go	Go	- Purchasing pre-manufactured components increases total project cost. - Cost may not justify marginal traction improvement over lower-cost alternatives.
3	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Maybe	Go	- 3D printed materials may wear down under continuous friction and load. - Rubber bands can stretch or degrade, reducing long-term reliability.
4	4 Wheel Drive	Go	N/A	Go	No Go	Go	Go	No Go	Go	Go	Go	- Requires synchronization between two ESP32 boards, increasing software complexity. - Coordinating encoder feedback and motor control adds unnecessary programming overhead.
5	2 Wheel Drive	Go	N/A	Go	Go	Maybe	Maybe	Go	Go	Go	Go	- Uneven weight distribution may reduce traction and stability. - Limited drive contact points can affect load handling performance.
6	Conveyor Belt + 2 Lane Knocking System	No Go	Go	Go	Go	Go	Maybe	Go	Maybe	Maybe	Maybe	- System footprint exceeds locker size constraints. - Additional belt and lane structure increases spatial requirements.
7	Steep Slide with Hatch System	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	- Sorting wheel timing must align precisely with item arrival. - Rotational misalignment may reduce throughput efficiency.
8	Slide With Sorting Wheel	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	
9	Slide	Go	N/A	N/A	Go	Go	Go	Go	Go	Go	Go	
10	Gate	Go	N/A	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Servo actuation introduces delay during opening and closing cycles. - Mechanical lag may slow repeated sorting operations.
11	UltraSonic Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
12	Infrared Sensor	Go	Go	No Go	Go	Go	Go	Go	Maybe	Go	N/A	- Lab IR sensor is unreliable for accurate distance measurement. - Readings are affected by ambient light and surface reflectivity.
13	UltraSonic + Infrared Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
14	Scoop pick-up	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	
15	Claw	Go	Go	N/A	Go	Maybe	Maybe	Go	Go	Maybe	Maybe	- Requires precise servo calibration and grip control. - Complex geometry increases fabrication difficulty.
16	Scoop pick-up + Gate	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Combining scoop and gate may introduce redundant handling steps. - Additional mechanism coordination increases cycle time.

Pugh Matrix for Normalizing Weights:

Objectives	Simplicity of Design	Sorting Accuracy & Amount	Efficient Weight Distribution	Navigation Accuracy	Efficient UI	Budget	Total	Normalized Weighting	Legend	
Simplicity of Design	1.00	0.14	1.00	0.20	9.00	0.14	11.49	0.12	Relative Importance	Score
Sorting Accuracy & Amount	7.00	1.00	7.00	1.00	9.00	1.00	26.00	0.27	Exceptionally Less	1/9
Efficient Weight Distribution	1.00	0.14	1.00	0.20	9.00	0.33	11.68	0.12	Much Less	1/7
Navigation Accuracy	5.00	1.00	5.00	1.00	9.00	0.20	21.20	0.22	Less	1/5
Efficient UI	1/9	1/9	1/9	1/9	1.00	0.11	1.56	0.02	Slightly Less	1/3
Budget	7.00	1.00	3.00	5.00	9.00	1.00	26.00	0.27	Equally Important	1.00
							97.92	1.00	Slightly More	3.00
									More	5.00
									Much More	7.00
									Exceptionally More	9.00

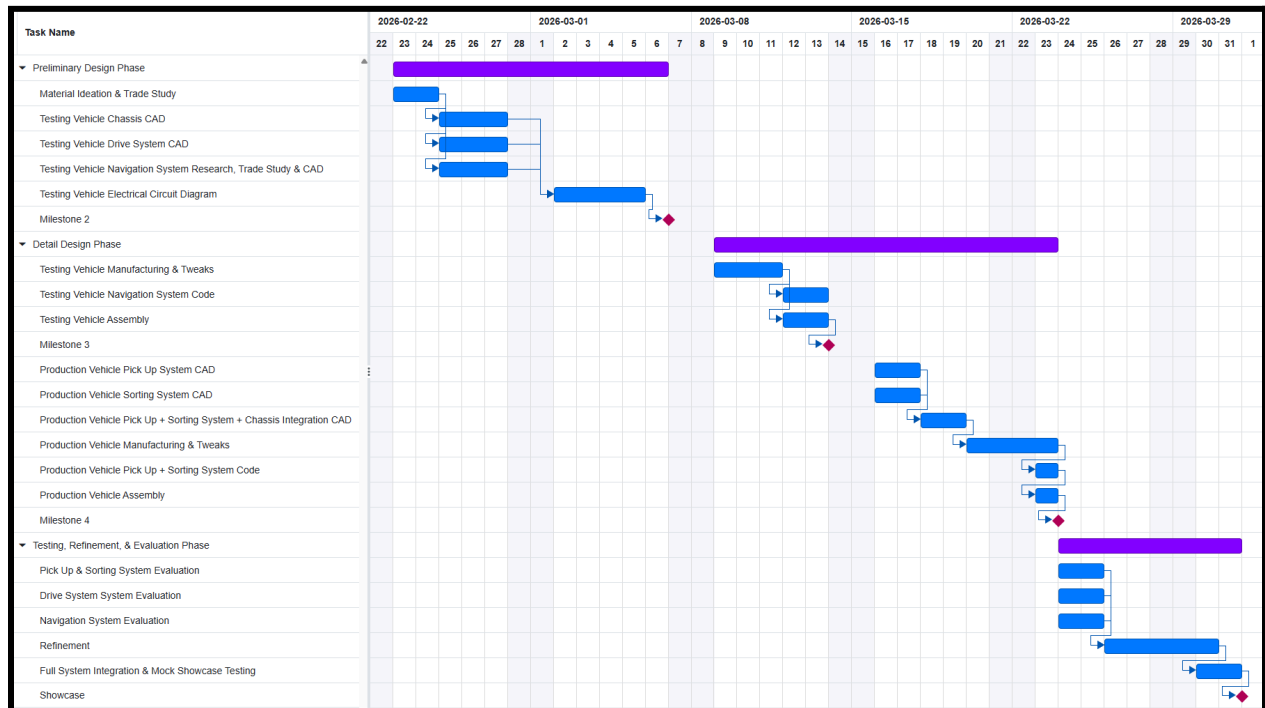
#	Objective	How to Measure	Weight (higher number is more important)	Gears with Tracks	Aftermarket Rubber Wheels	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	2 Wheel Drive	Steep Slide with Hatch System	Slide With Sorting Wheel	Slide	Gate	UltraSonic Sensor	UltraSonic + Infrared Sensor	Scoop pick-up	Claw
1	Simplicity of Design		0.12	-1	2	1	1	1	2	2	-1	0	-2	2	-1
2	Sorting Accuracy & Amount		0.27	0	0	0	2	2	2	0	0	0	0	0	0
3	Efficient Weight Distribution		0.12	2	1	1	0	-1	0	0	-1	0	0	-1	-1
4	Navigation Accuracy		0.22	-1	1	1	2	0	0	0	0	0	1	1	-1
5	Efficient UI		0.02	0	0	0	0	0	0	0	0	0	2	1	0
6	Budget		0.27	-1	-1	2	2	1	2	2	1	2	2	2	-1
Results															
Weighted Totals				-0.360848139	0.304824277	0.984119609	1.012416779	0.794644015	1.296718971	0.705059447	0.0289845675	0.5310595254	0.9447411787	0.8708069526	-0.7185838413

CONCEPTUAL DESIGN PHASE

Task Completed March 8, 2026

- Vehicle Material Trade Study Completed
- Test Vehicle CAD Chassis Design Established
- Test Vehicle BOM Established
- Test Vehicle Electrical Circuit Diagram Established
- Test Vehicle Navigation System Established
- Project Time Line Established

Project Time Line:



Material Trade Study:

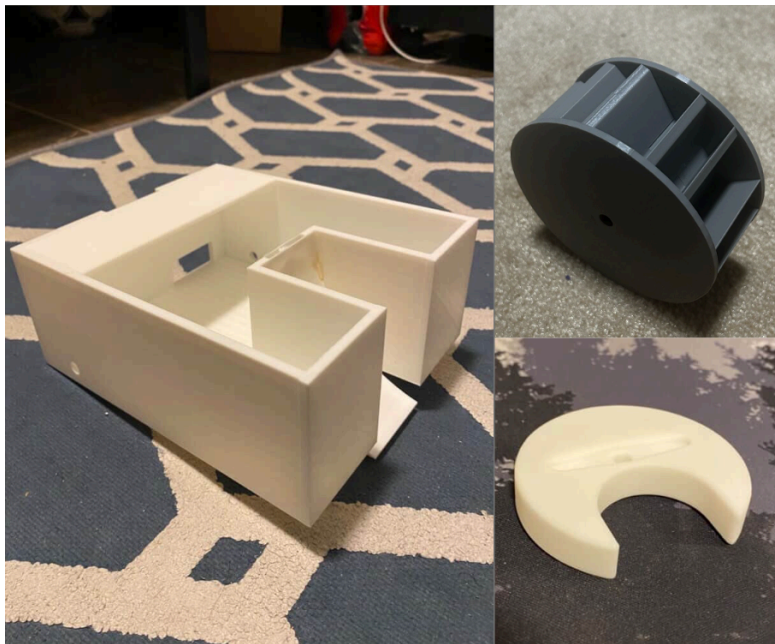
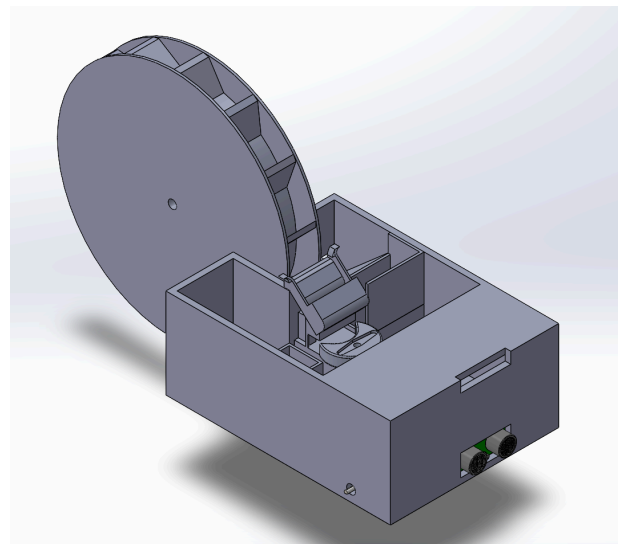
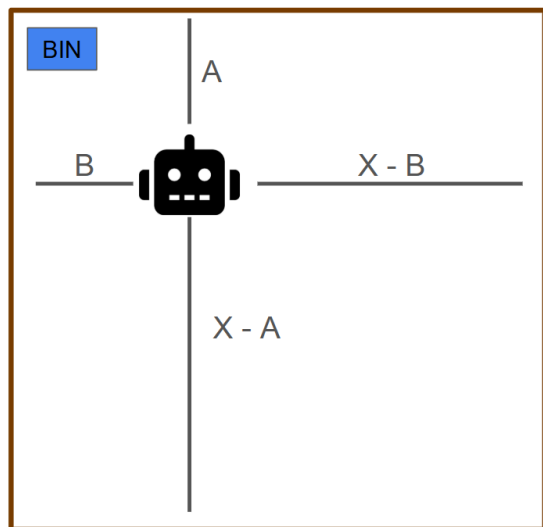
Metric	Weight	Lazer Cut Cardboard	3D Printed	Handcrafted Cardboard
Manufacturing Complexity	25.00%	2	1	-2
Weight	15.00%	2	-1	1
Design Complexity	30.00%	-2	0	1
Reliability	20.00%	-1	2	-2
Cost	10.00%	1	-1	2
Total	100.00%	0.1	0.4	-0.25

A big driver for using 3D printed materials instead of cardboard is material and equipment availability as 2 members of the group own 3D printers and filaments.

Navigation System Trade Study:

Metric	Weight	IR Beacon	Ultrasonic Sensor + Wall	IR Beacon + Ultrasonic Sensor + Wall
Reliability	40.00%	-1	1	2
Software Complexity	50.00%	-1	2	-2
Integration Complexity	10.00%	1	0	-1
Total	100.00%	-0.8	1.4	-0.3

Diagram of how navigation system will work:



Lessons Learned and Updated Design Goals

- Serviceability and Maintainability

- The design shall prioritize ease of maintenance by ensuring that frequently failing or high-wear components can be accessed, removed, and replaced without full system disassembly.
- Critical electronic components shall remain accessible in the fully assembled configuration, minimizing downtime during troubleshooting or repair.
- Modular design principles should be implemented, allowing individual subsystems to be serviced independently.
- Ease of Assembly and Disassembly
 - The product shall be designed for straightforward assembly using standard tools and minimal steps.
 - Components, particularly 3D printed parts, shall be designed as modular subassemblies that can be repeatedly assembled and disassembled without degradation.
 - Fastening methods should promote repeatability and alignment, reducing assembly errors and build time.
- Minimization of 3D Printing Supports
 - All 3D printed components shall be designed to minimize or eliminate the need for support structures, reducing material usage, print time, and post-processing effort.
 - Geometries should incorporate self-supporting angles ($\geq 30^\circ$ where possible) and avoid overhangs that require excessive support.
- Print Orientation and Structural Integrity
 - Parts shall be designed to allow optimal print orientation, ensuring that layer adhesion does not compromise structural performance.
 - Load-bearing components must be oriented such that principal stresses are aligned with the strongest print direction (layer plane).
 - Design geometry should account for anisotropic material properties inherent to FDM printing.
- Standardization of Fasteners
 - The design shall utilize M3 screw wherever possible to simplify manufacturing, assembly, and maintenance.
 - Use of non-standard fasteners shall be avoided unless functionally justified (e.g., load requirements, space constraints).
- Dimensional Tolerancing
 - General dimensional tolerances shall be maintained at ± 0.5 mm to ensure compatibility between interfacing components while accommodating typical manufacturing variability.
 - Fits between mating parts (e.g., press-fit, clearance fit) shall be designed with consideration of printer accuracy and material behavior.

Lessons Learned and Updated Constraints

- Manufacturing Capability Constraint (3D Printing)
 - The design shall not require dimensional tolerances tighter than ± 0.1 mm, reflecting the practical limitations of the Bambu Lab P1S 3D printer.
 - Features requiring higher precision shall be redesigned, simplified, or post-processed rather than relying on print accuracy alone.

- All critical dimensions must be validated against realistic printer performance, including variability due to material, calibration, and environmental conditions.

DETAIL DESIGN PHASE

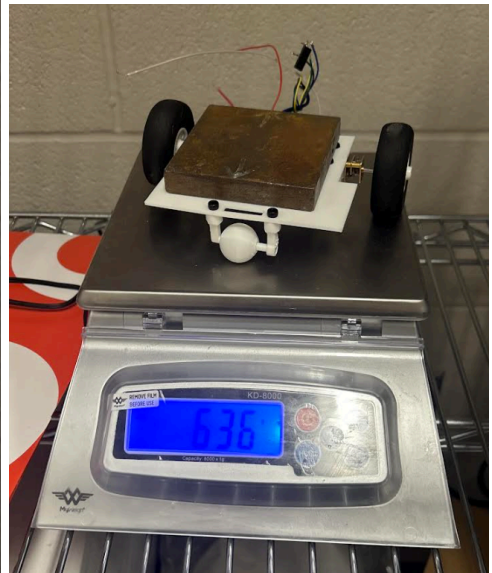
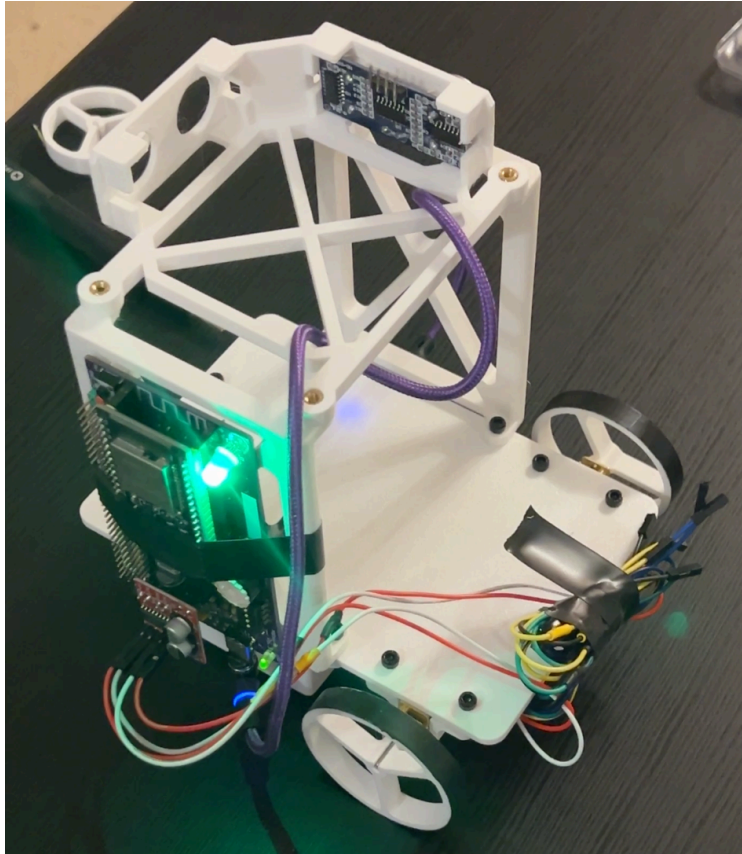
Task Completed March 13, 2026

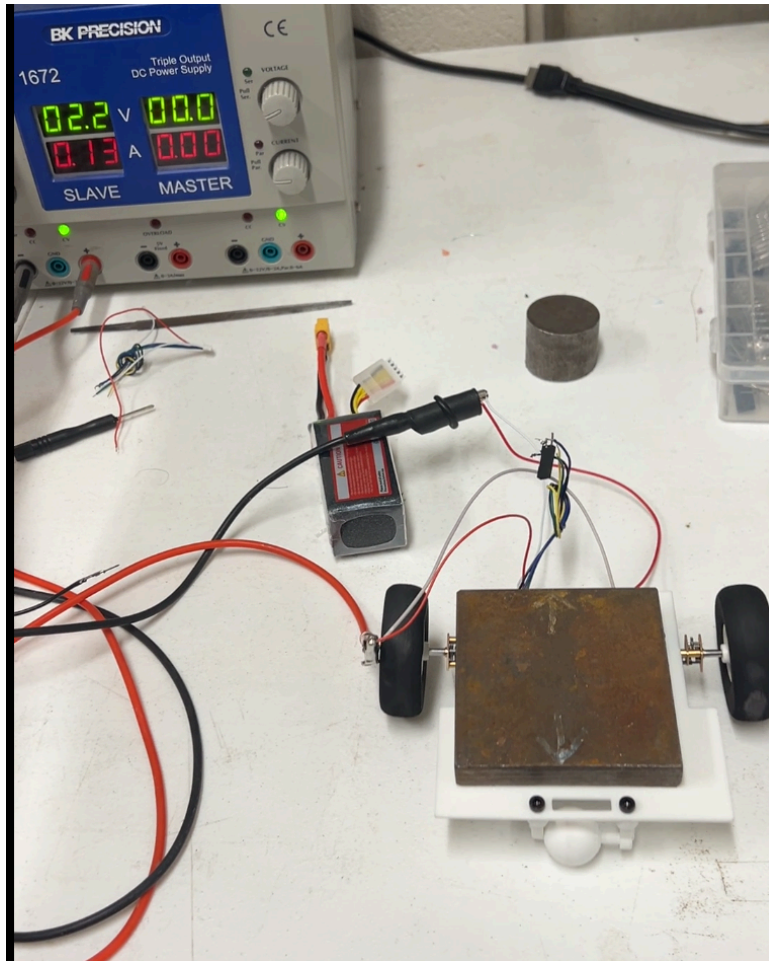
- Updated Project Objectives
- Updated Project Constraints
- Test Vehicle Completed

ID	Objectives	How to Measure
O-1	Simplicity of Design	The robot should be made with a minimum no. of components to increase reliability and durability.
O-2	Sorting Accuracy & Amount	Sort accuracy (the sorting between high-value and low-value erasers) should be > 85%.
O-3	Efficient Weight Distribution	The robot should be balanced and must not stumble under its own weight and be as light as possible.
O-4	Navigation Accuracy	Should traverse the operation without losing sight of goals and have at least 90% success rate on test runs.
O-5	Efficient UI	Clear indication of each process step and a failsafe in case of incidents or wrong performance of the task
O-6	Budget	The total cost of external components should be < \$20 provided by the project.
O-7	Design for Serviceability	The design shall prioritize ease of maintenance by ensuring that frequently failing or high-wear components can be accessed, removed, and replaced without full system disassembly. Critical electronic components shall remain accessible in the fully assembled configuration, minimizing downtime during troubleshooting or repair. Modular design principles should be implemented, allowing individual subsystems to be serviced independently.
O-8	Design for Assembly	The product shall be designed for straightforward assembly using standard tools and minimal steps. Components, particularly 3D printed parts, shall be designed as modular subassemblies that can be repeatedly assembled and disassembled without degradation. Fastening methods should promote repeatability and alignment, reducing assembly errors and build time.

O-9	Standardization	The design shall utilize M3 screw with M3 5 x 4 mm heated inserts wherever possible to simplify manufacturing, assembly, and maintenance. Use of non-standard fasteners shall be avoided unless functionally justified (e.g., load requirements, space constraints). General dimensional tolerances shall be maintained at ± 0.5 mm to ensure compatibility between interfacing components while accommodating typical manufacturing variability.
O-10	Speed	The amount of time per mission should be minimized < 1 minute.

ID	Constraint	Description
C-1	Size	The robot must be able to fit in the MSE lab locker.
C-2	Sorting of Erasers	The picked-up erasures must be sorted either on the robot or on a stationary sorting system, and must be done based on wavelength (colour).
C-3	Navigation	The robot must be able to traverse the area of operations without any hindrance or help from external sources.
C-4	Power/Battery	The robot must be able to perform its entire task on a single charge of the batteries used, and the batteries used must not be modified.
C-5	Weight Drivable	The robot must be able to move on its own weight and be light enough to be carried by a human (< 3 kg).
C-6	Time-Efficient	Perform the entire operation within the limit of 2 minutes.
C-7	Autonomous	Operations must be performed without any human intervention or wireless communication.
C-8	Budget	The overall cost of components for the robot, excluding components provided by the MSE 2202 lab kit, must not exceed \$50. The overall cost for FDM and laser cutting must not exceed \$20.
C-9	Durability	The robot must be able to perform its task without the expense of its components. The robot must be able to perform the entire operation > 50 times without any need for component replacement.
C-10	Manufacturability	The robot must be manufactured using available materials/components. This includes 4x MSE 2202 lab kit, FDM printed ABS/PETG/PLA components, and external components adhering to the budget constraint.

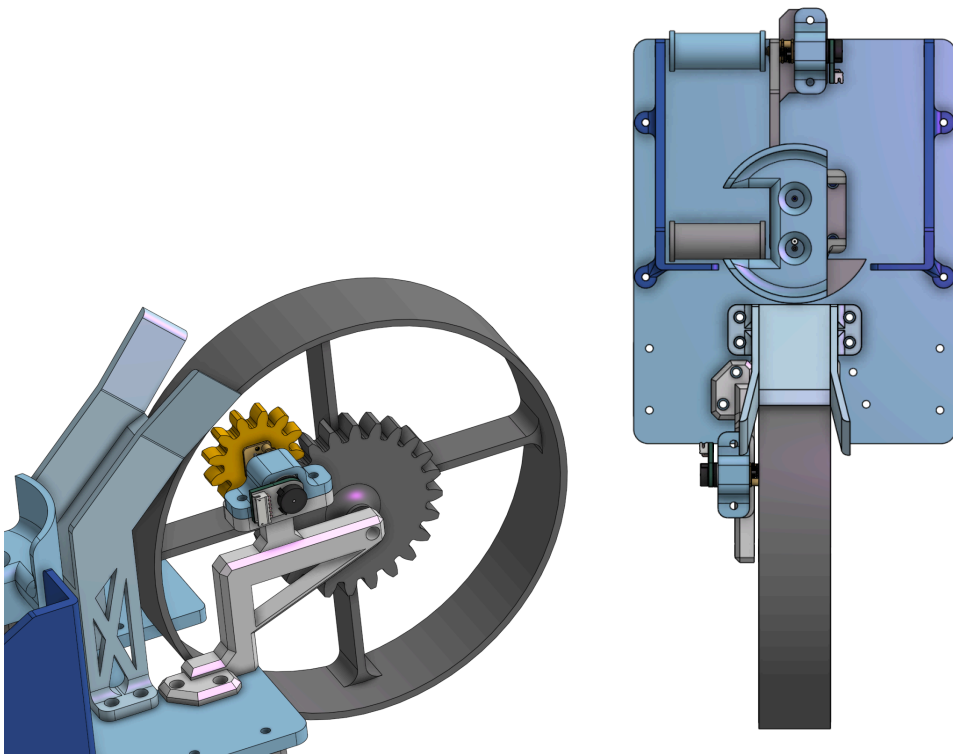
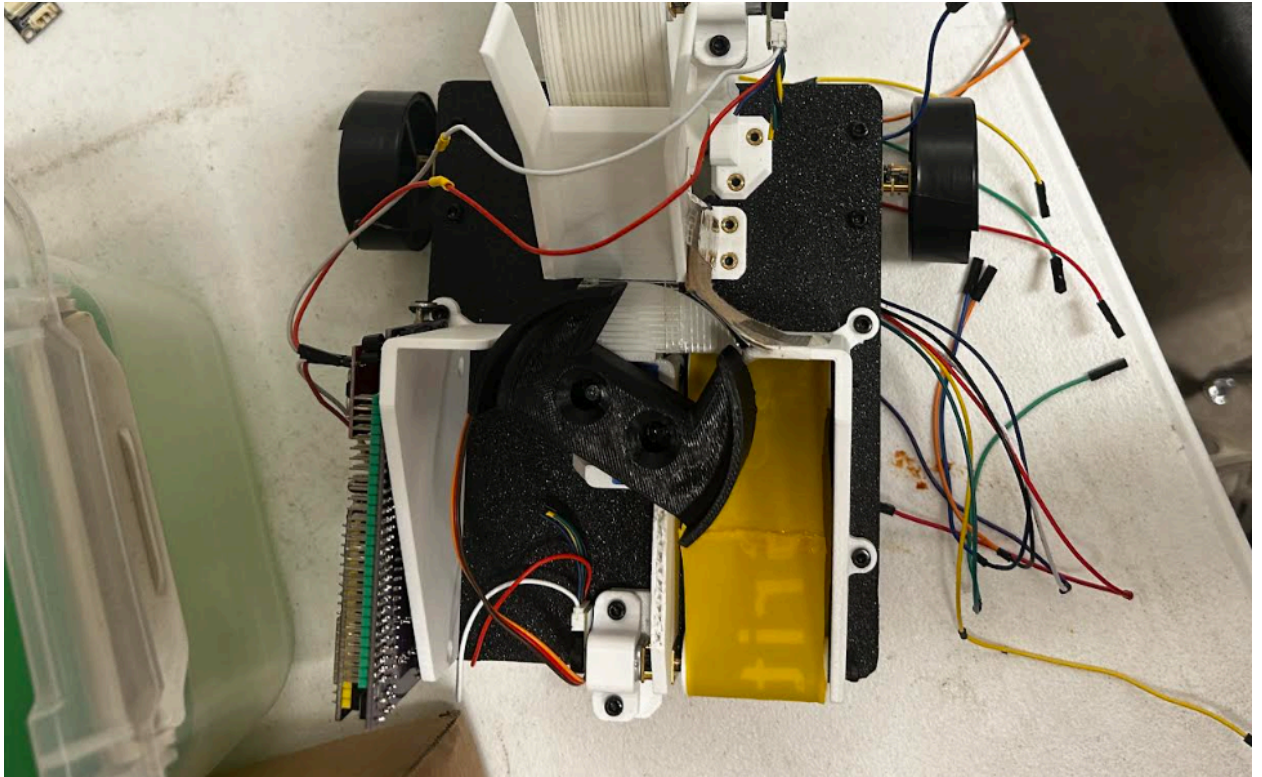




Task Completed March 24, 2026

- Tested Multiple Pickup System Implementation

- Implemented Sorting System & Integrated Subsystems



PRODUCT DESIGN PHASE

Task Completed March 31, 2026

- Implemented final pick up system
- Implemented Final Drive system
- Final integration achieved
- Mock mission testing completed

